



Semantic Processing During Spoken Word Production by Children with Cochlear Implants

Man Wang¹, Yixin Ding¹, Niels O. Schiller²

¹ School of Foreign Languages, Qingdao University, Qingdao, China

² Department of Linguistics and Translation, City University of Hong Kong, Hong Kong SAR
manwang@qdu.edu.cn, iseny_yixind@163.com, niels.schiller@cityu.edu.hk

Abstract

Research that investigates the speech production of children with cochlear implants (CIs) mostly focuses on the characteristics of their speech sounds. Few studies have looked at the psycholinguistic processes during speech production of children with CI. Our study examines the semantic processing during speech production of this group of speakers, compared to their normal hearing (NH) peers. Using the picture-word interference paradigm, we manipulated the semantic relatedness between target picture names and distractor words. We observed the typical semantic interference effect in the NH group but not in the CI group, suggesting that the semantic network may be organized differently in the CI group than in their NH peers and the CI group may have difficulties accessing the semantic categories. Furthermore, our results are in line with the suggestion that the CI group may rely more on a top-down strategy or attentional cognitive processing than a bottom-up semantic activation.

Index Terms: speech production, semantics, children with cochlear implants, psycholinguistics

1. Introduction

1.1. Semantic Processing of Children with CIs

Hearing impairment is one of the main causes of spoken language delay and deficits in children's development of language skills [1]. Cochlear implants (CIs), especially early CI implantation, can largely facilitate children with language impairments to develop language skills comparable to their normal hearing (NH) peers. Nevertheless, the hearing impairment before the CI implantation causes less optimal hearing and will still prevent children with CIs from reaching age-equivalent language capabilities resulting in deviance from their NH peers [2].

Empirical studies have made use of behavioral measurements of language development in children with CIs post implantation, and most examinations employ batteries of language tests (e.g., Peabody Picture Vocabulary Test, PPTV, Mean Length of Utterance, MLU) [3, 4]. Although these standardized tests are effective in examining the language capabilities of children with CIs, they are not sufficient to reveal the reasons behind the characteristics of the language development of children with CIs. For instance, battery tests show children with CIs are delayed in vocabulary development [5], but it is unknown whether the poor lexical representation or the phonological deficit attributes to the delay [6].

Attempts have been made to investigate the underlying mechanisms of language production in children with CIs and their findings suggest a deficit in semantic network development. For instance, children with CIs performed poorer in semantic verbal fluency task (e.g., to name as many objects as possible within a time limit) compared to the age- and IQ-matched group of NH peers, and were less efficient in accessing semantic categories than NH peers [7]. Still, their poorer performances could be attributed either to a poorer semantic representation development or to a deficit accessing the semantic representation during language production.

In the attempt to tease apart these two possibilities, evidence for a poorer semantic representation development has been found. By using a timed verbal fluency task, correlation and network analyses showed that children with CIs seemed to have a less-developed semantic network structure compared to their NH peers [6].

Later studies have made use of electrophysiological measurements to investigate semantic processing during spoken language processing [8]. These studies specifically looked at the amplitude and latency the event-related potential (ERP) component N400, which usually reflects semantic processing [9]. While the study with auditory stimuli found the typical N400 effect in the CI group at 18 and 24 months after implantation comparable to the NH group [10], studies with visual stimuli usually observed stronger N400 effects, despite lower semantic abilities, in the CI group compared to the NH group (see [8] for a review). Additionally, ERP differences were observed in an earlier P2 time window between the CI and NH groups, suggesting differences in the attentional and/or strategic processing that may impact semantic processing between groups. As suggested by [8], children with CIs may rely more on top-down processing as a compensatory strategy than on bottom-up language processing, therefore showing stronger N400 effects.

Although great progress has been made in the investigation of semantic processing in children with CIs, the implications are limited due to the small number of samples and limited studies that focus on the mechanisms of semantic processing instead of semantic performance. In addition, studies that use experimental tasks usually look at semantic processing during language comprehension process instead of language production.

1.2. Semantic Processing in Language Production of Children with CIs

A speaker needs to go through several major stages before successfully articulating a single word. Different psycholinguistics models of speech production generally agree

on the following stages of speech production: conceptual preparation where semantic processing takes place, lemma level where the lexico-syntactic information is processed, word-form encoding level where the phonological form of the word is encoded, and finally articulation (e.g., WEAVER++ model) [11, 12].

To study semantic processing, the picture-word interference paradigm has been widely used. In this paradigm, a participant is asked to name a target picture with a superimposed distractor on the picture. If the distractor (e.g., dog) is semantically related to the target picture (e.g., cat), it usually interferes the picture naming resulting in longer naming latencies, compared to an unrelated condition [14, 15]. The semantic relatedness here refers to belonging to the same semantic category. The semantic interference effect observed with the paradigm is considered to reflect the competition during lexical selection [16]. This is based on the assumption that words that belong to the same semantic category are connected stronger to each other in the semantic network and will be activated if neighbors in the network are activated.

Previous research raises the question whether the lower semantic abilities of children with CIs are attributed to the underdeveloped semantic network or the deficit in accessing the semantic representation. The picture word interference paradigm can be a valuable tool to investigate this issue. Nevertheless, studies using this paradigm seem to show discrepant findings. Either no significant semantic interference was observed [17, 18, 19] or greater inhibition was shown with the CI group compared to a NH group (see a preliminary study [7]). Please note that in these former two studies where no significant semantic effect was observed, the distractor was presented auditorily, which may not have been salient enough to reveal any potential semantic effect.

Therefore, the present study aims to investigate the semantic processing during speech production of children with CIs, in comparison with their NH peers. Specifically, by using the picture word interference paradigm, we examine semantic representations of words and semantic access in children with CIs. If the semantic network is developed equally well compared to their NH peers, a semantic interference effect is expected. If the semantic network is underdeveloped or semantic categories are not similarly formed compared to their NH peers, we may not observe any semantic interference in the CI group.

2. Methods

2.1. Participants

A total of 20 children with hearing impairments (11 male, mean age = 10.9 years, SD = 3.24) from special education schools and 20 normally hearing children (8 male, mean age = 10.1 years, SD = 2) participated in this experiment. All hearing-impaired children were pre-lingually deaf and had a cochlear implant before the age of three. They were all right-handed and had normal or corrected-to-normal vision and no history of neurological impairment. All participants were paid for their participation.

These 40 children performed a test for nonverbal intelligence (Combined Raven's Test; CRT), which provided a global estimation of their intelligence. From this test, we could conclude that the children with CIs performed within the range of normal intelligence. The two groups did not differ significantly in their CRT scores, $t(38) = 0.742$, $p > .05$.

Audiological screening was conducted for hearing-impaired children. Based on a questionnaire, participants who experienced prelingual hearing loss and those with a hearing loss of 70 dB or above were selected.

2.2. Materials and Design

Thirty black-and-white line drawings of common objects were selected from a standardized picture database [20]. All these pictures were combined with distractors. The distractors were selected based on their congruency with the target picture names in terms of semantic category. The factor had two levels: semantically related versus semantically unrelated. On each trial, pictures were presented with a distractor (from one of the two conditions) superimposed on the center of the picture. The distractors were not phonologically or orthographically related to the target picture names. The distractors in the two conditions were comparable in terms of word frequency (calculated with the log frequency of words in the SUBTLEX-CH database) [21] and visual complexity (number of strokes), $t_s < 2$, $p_s > .05$.

Since we are interested in semantic processing, we choose a negative SOA value (i.e., -100 ms) where the semantic interference effect was widely observed in previous studies [22,23].

Table 1: An example of a target picture presented with distractor in each condition. Distractors either match or mismatch the semantic category of target picture name.

Target Picture Name	Distractor Condition	
	Semantically related	Semantically unrelated
兔子 /tu4zi0/ rabbit	刺猬 /ci4wei4/ hedgehog	照片 /zhao4pian4/ photo

2.3. Procedure

The experiment was administered on the psychology software tool PsychoPy (v2023.1.3) on a Lenovo Thinkbook laptop. Participants were tested individually in a quiet room, where they were seated in front of a monitor at a distance of approximate 60 cm. They were asked to name the pictures as fast and as accurately as possible. Vocal response times were measured by a voice key connected to a microphone installed on the laptop.

The experiment consisted of three sessions: a familiarization session, a practice session and an experimental session.

In the familiarization session, each picture was presented once with its name underneath. Participants were requested to view the images and their names.

In the practice session, each picture was presented once with "XX" superimposed on it and participants were asked to name the pictures with the correct names while ignoring the "XX" on the pictures. Responses that deviated from the names given in the familiarization session were corrected by the experimenter.

In the experimental session, participants saw each of the 30 pictures twice based on different combination of semantic conditions, resulting in 60 experimental trials per participant, which were presented in a pseudo-random order such that no

two consecutive trials were from the same distractor condition. All trials were divided equally into 4 blocks with a short break in between (the length of the break was determined by the participant). On each trial, a fixation point (“+”) was presented for 300 ms, followed by a blank screen (200 ms), the target picture with distractor (displayed until the participant initiated a vocal response, with a 3 s time-out), followed by another blank screen (2 s) before the next trial began (Figure 1).

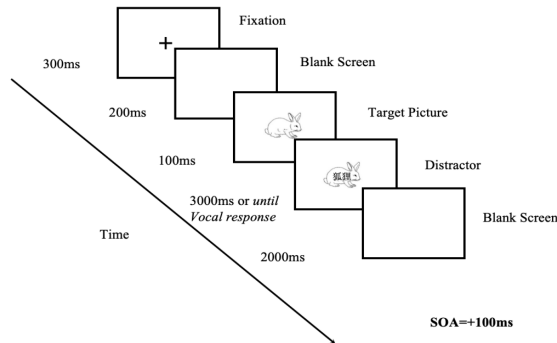


Figure 1: Trial sequence in the experiment

3. Results

3.1. The CI Group

For the CI group, 17.75% of all data points (2,400) comprising: (a) incorrect responses; (b) voice-key failures (the first two types were counted as errors; the error rate was 16.46%); (c) outliers (i.e., naming latencies exceeding 3 SDs above or below the participant's mean; 3.75%) were removed from the behavioral data analysis.

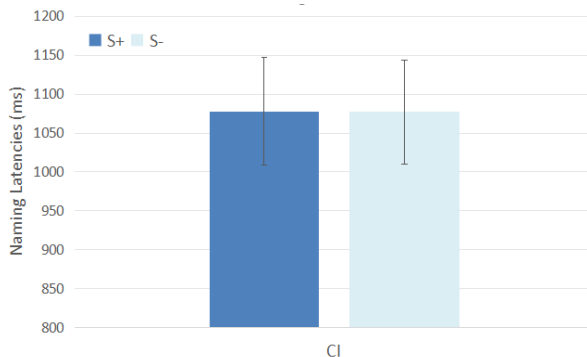


Figure 2: The mean naming latencies in the semantically related (S+) and unrelated (S-) conditions of the CI group. The error bars indicate the standard errors (SE) of each condition.

A paired-sample t-test was conducted to compare the naming latencies between the semantically related and unrelated conditions. The naming latencies for the related condition (mean = 1,078 ms, SD = 307 ms) were not significantly different from the unrelated condition (mean = 1,078 ms, SD = 300 ms), $t(19) = .030$, $p > .05$. The results suggest that in the group of children with CIs, there is no significant semantic interference effect during their speech production with the picture-word interference paradigm (Figure 2).

The error rates of the two distractor conditions were also submitted to paired-sample t-tests, and the participants in the CI

group did not show significant differences between the semantically related (mean = 17.08%, SD = 7.19%) and unrelated conditions (mean = 18.42%, SD = 7.08%).

3.2. The NH Group

For the NH group, 15.29% of all data points (2,400), comprising: (a) incorrect responses; (b) voice-key failures (the first two types were counted as errors; the error rate was 14.17%); (c) outliers (i.e., naming latencies exceeding 3 SDs above or below the participant's mean; 5.17%) were removed from the behavioral data analysis.

A paired-sample t-test was conducted to compare the naming latencies between the semantically related and unrelated conditions. The naming latencies for the related condition (mean = 1,333 ms, SD = 249 ms) were significantly slower than those from the unrelated condition (mean = 1,256 ms, SD = 222 ms), $t(19) = 3.98$, $p = .001$ (two-tailed). The results suggest that in the NH group, there was a significant semantic interference effect during their speech production with the picture-word interference paradigm (Figure 3).

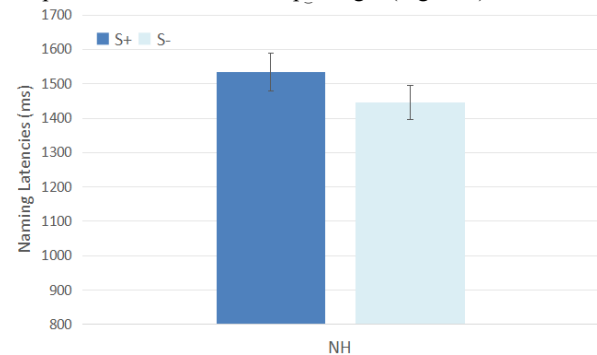


Figure 3: The mean naming latencies in the semantically related (S+) and unrelated (S-) conditions of the NH group. The error bars indicate the standard errors (SE) of each condition.

The error rates were also submitted to paired-sample t-tests, and the participants in the NH group seemed to make more errors in the semantically related condition (mean = 16.76%, SD = 8.54%) than in the semantically unrelated condition (mean = 13.83%, SD = 8.22%), but the difference did not reach significance, $t(19) = 1.835$, $p = .082$ (two-tailed).

3.3. Post-hoc Analysis on the Group Difference

The mean naming latencies of the two participant groups were submitted to independent-sample t-test as a post-hoc analysis to compare the group differences. The CI group generally responded faster (mean = 1,078 ms, SD = 301 ms) than the NH group (mean = 1,294 ms, SD = 232 ms), and the difference was significant, $t(38) = -2.55$, $p = .015$ (two-tailed).

The error rates across the two groups were also submitted to an independent-sample t-test, but the difference between the CI group (mean = 17.75%, SD = 6.65%) and the NH group (mean = 15.29%, SD = 7.59%) was not significant, $t(38) = 1.09$, $p > .05$.

4. Discussion

The present study investigated the semantic processing during speech production in children with cochlear implants (CIs)

compared to their normal hearing (NH) peers using the picture-word interference paradigm. Results revealed a significant semantic interference effect in the NH group, where naming latencies were slower in the semantically related condition compared to the unrelated condition. However, no such effect was observed in the CI group, suggesting differences in the underlying mechanisms of semantic processing during speech production between the two groups.

4.1. Semantic Processing in Children with CIs in Comparison with NH Peers

As introduced in the Introduction, the poorer performances of children with CIs in semantic tasks could either be attributed to the underdevelopment of semantic networks or the poorer ability to access the semantic representations of words. The absence of a semantic interference effect in the CI group is more likely to align with previous research indicating that children with CIs may have underdeveloped semantic networks [7]. This is because the semantic interference effect is believed to be caused by the competition of semantically related words (i.e., the distractor) that are close to the target word in the semantic network at the stage of lexical retrieval [16]. The lack of interference could be attributed to weaker connections within the semantic network, which may not activate competing lexical items to the same extent as in NH children.

Based on post-hoc analysis results that the CI group's naming latencies were generally faster than those of the NH group while the error rates were comparable, it seemed that children with CIs could access the semantic representations of the target words as successfully and accurately as, and even faster than their NH peers. This finding supports the hypothesis that children with CIs may rely on different cognitive strategies, such as top-down processing, to compensate for deficits in bottom-up semantic activation [8]. In the current task, children with CIs may adopt more of the strategy to focus on the target picture and ignore the visually-presented distractor, and result in generally faster naming latencies and no semantic interference effect.

In contrast, the NH group exhibited the typical semantic interference effect, consistent with prior studies using the picture-word interference paradigm with healthy adults [22, 23]. The competition between semantically related lexical items during speech production reflects that children in the NH group have the semantic category knowledge and have developed semantic networks based on their semantic categories, at least of the tested words in the current study.

Furthermore, the NH group's naming latencies were generally slower than those of the CI group, could probably be attributed to the interference caused by the distractors. In addition, it could also reflect that compared to the CI group, the NH group may rely more on semantic activations of both target picture names and distractors, rather than putting an emphasis on the strategy to ignore the distractors.

4.2. Implications for Language Development

The findings have important implications for understanding the language development of children with CIs. The presence of this effect in the NH group but not in the CI group underscores the potential impact of early auditory deprivation on the development of semantic networks and lexical access. Specifically, while CIs provide access to auditory input, the quality and timing of this input may not fully compensate for the early auditory deprivation experienced by these children

prior to the CI implantation. This could lead to differences in how lexical-semantic information is organized and accessed during speech production [6]. The results suggest that interventions aimed at improving semantic category knowledge and strengthening semantic networks may be beneficial for children with CIs.

4.3. Limitations and Future Directions

While the results of the current study can provide implications for the underlying mechanisms for the deficit in semantic processing of children with CIs, it is still unclear how the deficit in semantic network development is affected, or to what extent children with CIs rely on strategic processing in comparison to semantic processing in speech production.

Therefore, studies with children with hearing impairments before CI implantation and longitudinal studies are needed to explore how semantic processing evolves over time in children with CIs and how early interventions might influence these developmental trajectories. Furthermore, additional tasks, such as semantic priming, verbal fluency, or cross-modal picture-word interference tasks, could provide a more comprehensive understanding of semantic processing in children with CIs.

5. Conclusion

In conclusion, the present study provides evidence that children with CIs may process semantic information differently during speech production compared to their NH peers. The absence of a semantic interference effect in the CI group suggests potential deficits in semantic network development, especially semantic category acquisition. These findings highlight the need for targeted interventions to support the language development of children with CIs and underscore the importance of further research into the cognitive and linguistic mechanisms underlying their speech production.

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7. References

- [1] A. Hogan, M. Shipley, L. Strazdins, A. Purcell, and E. Baker, "Communication and behavioral disorders among children with hearing loss increases risk of mental health disorders," *Australian and New Zealand Journal of Public Health*, vol. 35, no. 4, pp. 377–383, Aug. 2011.
- [2] B. Schramm, A. Bohnert, and A. Keilmann, "Auditory, speech and language development in young children with cochlear implants compared with children with normal hearing," *International Journal of Pediatric Otorhinolaryngology*, vol. 74, no. 7, pp. 812–819, Jul. 2010.
- [3] L. M. Dunn and L. M., *Peabody Picture Vocabulary Test (3rd edn)*. Circle Pines: American Guidance Service, 1997.
- [4] P. J. Blamey *et al.*, "Relationships among speech perception, production, language, hearing loss, and age in children with impaired hearing," *Journal of Speech, Language and Hearing Research*, vol. 44, no. 2, pp. 264–285, Apr. 2001.
- [5] C. Ouellet, M.-T. Le Normand, and H. Cohen, "Language evolution in children with cochlear implants," *Brain and Cognition*, vol. 46, no. 1–2, pp. 231–235, Jun. 2001.

- [6] Y. N. Kenett, D. Wechsler-Kashi, D. Y. Kenett, R. G. Schwartz, E. Ben-Jacob, and M. Faust, "Semantic organization in children with cochlear implants: computational analysis of verbal fluency," *Frontiers in Psychology*, vol. 4, no. 543, pp. 1-11, Sep. 2013.
- [7] R. G. Schwartz, S. Steinman, E. Ying, E. Y. Mystal, and D. M. Houston, "Language processing in children with cochlear implants: A preliminary report on lexical access for production and comprehension," *Clinical Linguistics & Phonetics*, vol. 27, no. 4, pp. 264-277, Mar. 2013.
- [8] P. Kallioinen, J. K. Olofsson, and C. Nakeva, "Semantic processing in children with Cochlear Implants: A review of current N400 studies and recommendations for future research," *Biological Psychology*, vol. 182, no. 108655, pp. 1-12, Aug. 2023.
- [9] M. Kutas, and K. D. Federmeier, "Thirty years and counting: finding meaning in the N400 component of the event-related brain potential (ERP)," *Annual Review of Psychology*, vol. 62, no. 1, pp. 621-647, Jan. 2021.
- [10] N. K. Vavatzanidis, D. Mürbe, A. D. Friederici, and A. Hahne, "Establishing a mental lexicon with cochlear implants: an ERP study with young children," *Scientific Reports*, vol. 8, no. 910, 1-13, Jan. 2018.
- [11] W. J. M. Levelt, A. Roelofs, and A. S. Meyer, "A theory of lexical access in speech production," *Behavioral and Brain Sciences*, vol. 22, no. 01, pp. 1-38, Feb. 1999.
- [12] A. Roelofs, "Self-Monitoring in Speaking: In Defense of a Comprehension-Based Account," *Journal of Cognition*, vol. 3, no. 1, pp. 1-13, Sep. 2020.
- [13] J. D. Jescheniak and H. Schriefers, "Discrete serial versus cascaded processing in lexical access in speech production: Further evidence from the coactivation of near-synonyms," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 24, no. 5, pp. 1256-1274, 1998.
- [14] H. Schriefers, A. S. Meyer, and W. J. M. Levelt, "Exploring the time course of lexical access in language production: Picture-word interference studies," *Journal of Memory and Language*, vol. 29, no. 1, pp. 86-102, Feb. 1990.
- [15] W. La Heij, "Components of Stroop-like interference in picture naming," *Memory & Cognition*, vol. 16, no. 5, pp. 400-410, Sep. 1988.
- [16] S. Jerger, L. Lai, and V. A. Marchman, "Picture Naming by Children with Hearing Loss: I. Effect of Semantically Related Auditory Distractors," *Journal of the American Academy of Audiology*, vol. 13, no. 09, pp. 463-477, Oct. 2002.
- [17] S. Jerger, L. Lai, and V. A. Marchman, "Picture Naming by Children with Hearing Loss: II. Effect of Phonologically Related Auditory Distractors," *Journal of the American Academy of Audiology*, vol. 13, no. 09, pp. 478-492, Oct. 2002.
- [18] B. E. de Hoog, M. C. Langereis, M. van Weerdenburg, H. E. T. Knoors, and L. Verhoeven, "Lexical access in children with hearing loss or language impairments, using the cross-modal picture-word interference paradigm," *Research in Developmental Disabilities*, vol. 37, pp. 81-94, Feb. 2015.
- [19] J. G. Snodgrass and M. Vanderwart, "A standardized set of 260 pictures: Norms for name agreement, image agreement, familiarity, and visual complexity," *Journal of Experimental Psychology: Human Learning and Memory*, vol. 6, no. 2, pp. 174-215, 1980.
- [20] Q. Cai and M. Brysbaert, "SUBTLEX-CH: Chinese Word and Character Frequencies Based on Film Subtitles," *PLoS ONE*, vol. 5, no. 6, e10729, Jun. 2010.
- [21] H. Schriefers, A. S. Meyer and W. J. M. Levelt, "Exploring the time course of lexical access in language production: Picture-word interference studies," *Journal of Memory and Language*, vol. 29, no. 1, pp. 86-102, Jan. 1990.
- [22] M. Wang, Y. Chen, M. Jiang, and N. O. Schiller, "The time course of speech production revisited: No early orthographic effect, even in Mandarin Chinese," *Language, Cognition and Neuroscience*, vol. 36, no. 1, pp. 13-24, 2021.